Date
Page

Chapter - 3

Computer software and operating system

1. Software

Software is the collection of programs, design to perform any task by giving instruction to the hardware resources.

It is a organized set of programs which controls, integrates and manages the hardware components.

Types of software

1) System software
2) Application software

System software

It is a collection of programs responsible for managing controlling the hardware devices of computer system. This software provides the environment for running other application software.

System software acts as an intermediate layer bet^n the user and computer hardware. A user interacts on commands, the hardware components to perform any work with the help of system software. System software is the main program that gets initialized and loaded in the computers primary memory continuously adding different programs and actions to run efficiently.

Some of the system software are: Operating system, Device Drivers, Language Translators, and Utility Programs.

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a. Operating System

It is a collection of program that manages and control the resources and overall activities of the computer system. It is the primary software that is responsible for providing general services to all other application programs. Ex: Windows, Linux, android etc.

b. language Translator

It is a set of programs that convert language written in high-level languages into lower machine level languages, so that computer can understand and work accordingly. Types of language translator are:-

i) Assembler

It converts program written in assembly language to machine language.

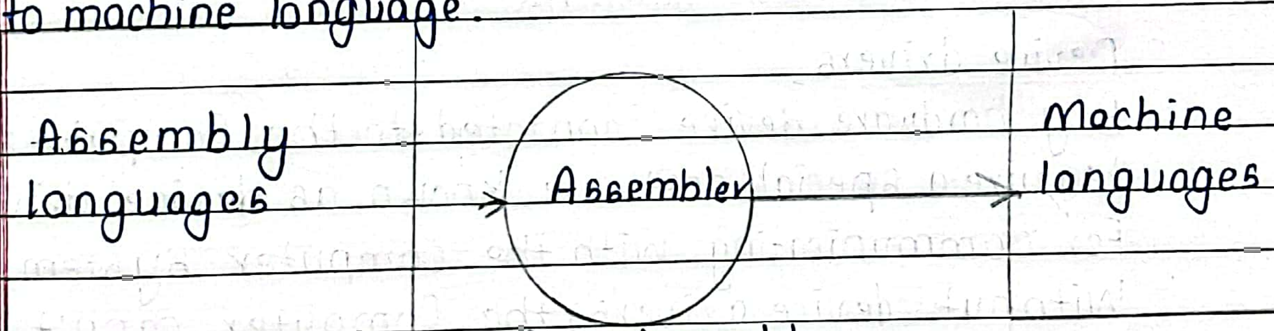


Fig: Assembler

ii) Interpreter

It is a language translator that translates one line of high-level language into machine level language at a time. Ex:- QBASIC, Pythen, Interpreter etc.

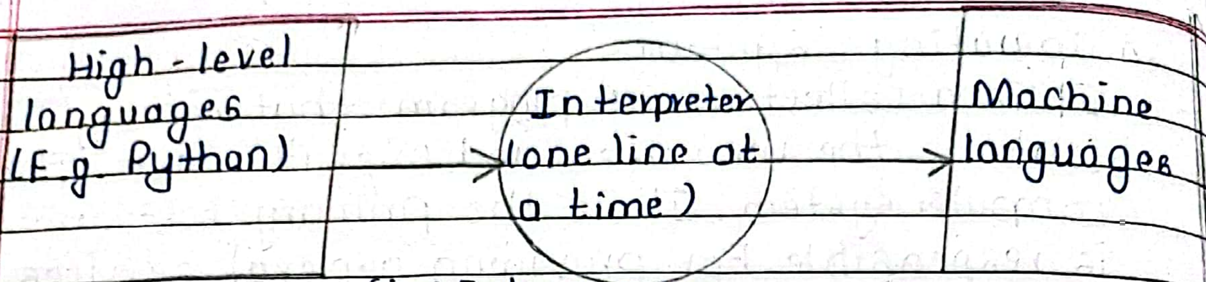


Fig: Interpreter

iii. Compiler

It is the language translator that translates all the statement written in high-level language into machine language in a single operation. Ex:- C, C++ compiler etc.

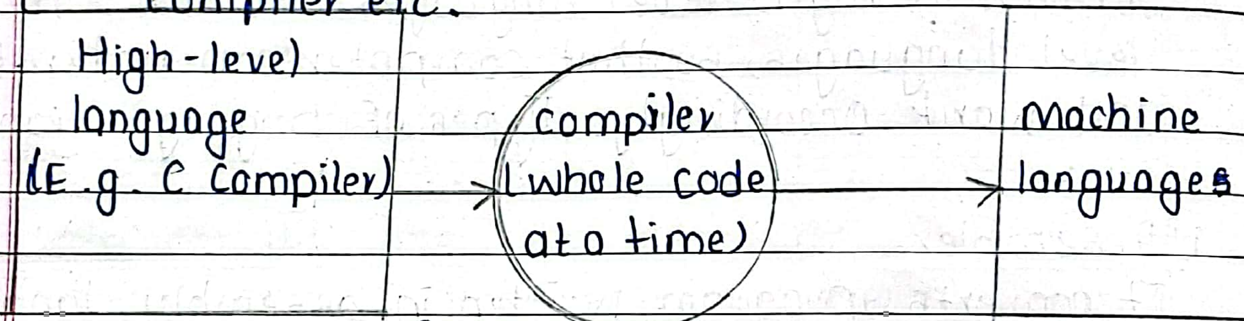


Fig: Compiler

c. Device drivers

Any hardware device connected to the Computer system require a special software known as device driver For communicating with the computer system. Without device drivers, the Computer cann't send and receive data correctly to a hardware devices such a printer

d. Utility Software

- Utility software helps to manage, maintain and control computer softwares.
- Ex:- antivirus software, backup software and disk tools.

2) Application Software

It is a collection of programs design to perform specific task. This software is also called enduser software because they are developed to be used by the endusers. Ex:- Ms-word, Excel, etc.

Types of application software

a. Packaged Software

- It is the most common type of software used in more than one environment.
- This software are developed to meet the requirement of large audience. For Ex:- Ms office package, is used by various users, schools, offices etc

b. Tailored Software

- The software which are developed undemand by the user is known as Tailored software.
- The tailored software varies according to the environment and users, who use it.
- For Ex:- School Management system is useful only to the concerned school rather than other hospitals.

Other types of software

1) Web-based software

The type of software that require a web browser to connect and access it's content is called Web-based software.

Date
Page

Advantages of Web-based Software
1) It can be assest from ~~only where~~ any where.
2) They can run on various devices like mobile, desktop etc.
3) Web-based software are easier to deploy and mountain.

2) Desktop Application.
- It is the program which are directly installed in a computer and does not require any connection to server for performing task.
- It is the program that runs locally in a desktop or personal computer.

3) Mobile Applications.
- The type of application software design to run on mobile devices is called mobile Application.
- Mobile Application helps users by connecting them to internet services and other services on a portable device.

Difference between System Software and Application Software

	System Software	Application Software
1.	It acts as an interface between the system hardware and running applications.	It is developed as per the requirement of the user
2)	It focuses on managing the whole system.	It focuses on solving a problem.
3)	A computer cann't run without system software	A computer can run without application software
4)	Designig and developing system software is complex process.	Designing and developing application software is relatively easy.
5)	Ex: Android OS, Disk cleanup software etc	Ex: MS Word, School Management software etc.

Difference between Package and Tailored software

	Packaged software	Tailored software
1)	This are the software design to solve common problems of the people.	This are the software design to solve specific problems of the people
2)	They are also called commercial software and are easily available for public users.	They are also called customized software and are available for personal use.
3)	They are less costly	They are expensive

Ex: MS Office packages	Ex: School Management software

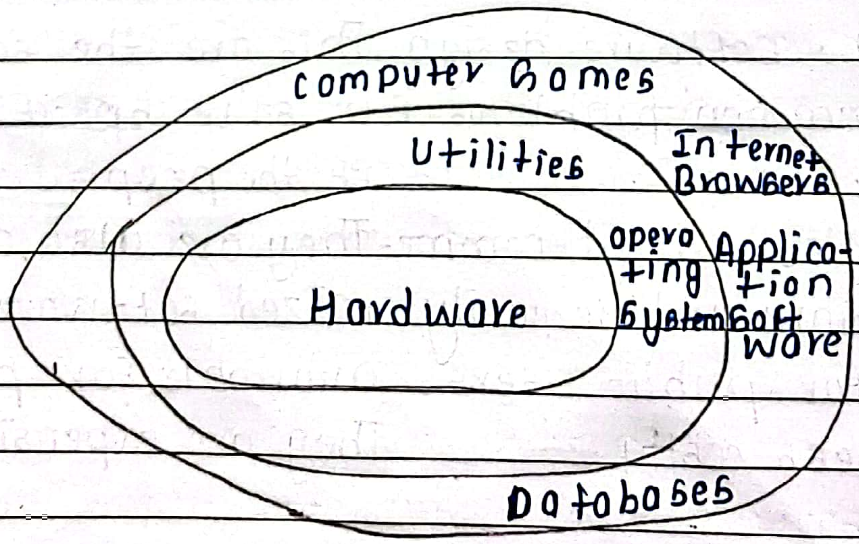
Chapter 3.2

Operating system

Concept of operating system

- The operating system is a collection of software and programs that manages computer hardware resources and provides services to the computer software
- It acts as an interface between the computer software and hardware.
- Operating system is a system software that manages the overall resources and activities of the computer system by allowing user to communicate with the hardware components.
- Ex:- Windows OS, Android OS, IOS, UNIX, LINUX etc.

B. Roles of operating System



Date
Page

i. OS ~~proc~~ creates and controls the file structure and directories.
ii. It provides the user interface.
iii. It hides the complexity of computer hardware.
iv. OS is responsible for providing security to the computer system.
v. The interrupt signals generated by I/O device and processor are handled by OS.
vi. It creates an environment for running application programs.

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Functions of operating system

a. File Management

While working on a computer a user needs to create, modify, rename and delete the files. Operating system allows the users to perform all this operations and to keep track of this files and their respective directors.

b. Process management

Process referce to the program under execution. OS is responsible for providing CPU with the currently executing processes and alocating required memory spaces to those processes.

c. Memory management

It refers to the management of primary memory mostly focused on RAM. OS is responsible for ~~Ep~~ allocating the memory to the runnings programs in the computer. Also it deallocates the memory after completion of process/programs.

d. Security

OS provides mechanism for securing the computer system along with the data and information stored in it by managing various users, and assigning the required roles to them. It ~~refers~~ further provides services of using passwords to add extra layers of security.

e. User interface

OS helps to communicates the user with the computer hardware. It hides the complicity of internal mechanisms and helps in the easy. intraction between the user and computer system.

F) Peripheral device management

Several input ~~do~~ and output devices are connected together in a computer system to per-

~~multiprogramming OS~~ form their task as operating system helps to manage this intraction by tracking all the peripheral and controlling them.

g) Deadlock Prevention

Deadlock is the condition where is process in a group is waiting for an event that only another process in a group can cause. Here each process is waiting for the release of some resources currently possesed by another process of the group. Here, none of the process can not release the resource hold by them. Because all the processor are waiting for another and none of them cause any events and wait remains forever, Operating system prevents the occurring of deadlock and manages if it occurs.

Types of Operating Systems

1 Single-User Operating System

A **single-user operating system** is designed to allow **only one user to use the computer at a time**. It manages the computer's hardware and software resources for that user.

Example: MS-DOS.

Scenario in daily life: A personal computer where one person logs in and uses the system.

2 Multiprogramming Operating System

A **multiprogramming operating system** allows **several programs to be kept in the main memory at the same time**. When one program is waiting for an input or output operation, the CPU can work on another program. This helps to keep the CPU busy and improves system efficiency.

Example: UNIX.

Scenario in daily life: If one program is waiting for a file to load, the CPU can process another program instead of remaining idle.

3 Multitasking Operating System

A **multitasking operating system** allows a user to **run and work with several applications at the same time**. The operating system quickly switches the CPU between different tasks, making them appear to run simultaneously.

Example: Windows 10/11, macOS.

Scenario in daily life: A user can listen to music, browse the internet, and type a document at the same time.

4 Multiprocessing Operating System

A **multiprocessing operating system** supports the use of **two or more processors or CPU cores** to perform tasks. Different processors or cores can execute different tasks simultaneously, which increases the speed and performance of the computer.

Example: Linux, Windows, and macOS running on multi-core computers.

Scenario in daily life: One CPU core can process a web browser while another core handles a video player at the same time.

5 Real-Time Operating System

A **Real-Time Operating System (RTOS)** is an operating system designed to respond to events or requests within a **specific time limit**. It is mainly used in systems where timely response is important.

There are two main types of real-time operating systems:

5.1 Hard Real-Time Operating System

A **Hard Real-Time Operating System** must complete a task within a strict deadline. Missing the deadline can cause serious problems or failure of the system.

Example: Airbag control systems, aircraft control systems, and medical equipment.

Scenario in daily life: An airbag system must respond immediately when a crash is detected. A delayed response could be dangerous.

5.2 Soft Real-Time Operating System

A **Soft Real-Time Operating System** also tries to complete tasks within a given time limit, but occasional delays are acceptable. Missing a deadline may reduce performance but usually does not cause complete system failure.

Example: Video streaming, online gaming, and multimedia systems.

Scenario in daily life: If a video frame is slightly delayed while watching a video, the video may become less smooth, but the system can continue working.

6 Distributed Operating System

A **distributed operating system** manages a group of independent computers and makes them work together as if they were a **single system**. The computers communicate with each other through a network and share resources and tasks.

Example: Distributed systems used in large computing networks and research environments.

Scenario in daily life: A large task can be divided among several connected computers. Each computer performs part of the task, and the results are combined to complete the overall task.

7 Difference Between Single-User, Multiprogramming, Multitasking, and Multiprocessing OS

Single-User OS	**Multiprogramming OS**	**Multitasking OS**	**Multiprocessing OS**
Allows one user to use the system at a time.	Keeps multiple programs in memory at the same time.	Allows a user to run multiple tasks at the same time.	Uses two or more processors or CPU cores.
Focuses mainly on supporting a single user.	Focuses on keeping the CPU busy by switching between programs.	Focuses on allowing multiple tasks to run at the same time.	Focuses on executing multiple tasks simultaneously.
Usually handles one user's activities.	The CPU switches to another program when one program is waiting.	The CPU rapidly switches between different tasks.	Different processors or cores can execute different tasks.
It may have limited resource-sharing capabilities.	Improves CPU utilization and reduces CPU idle time.	Improves user convenience and system responsiveness.	Improves performance and processing speed.
Example: MS-DOS	**Example:** UNIX	**Example:** Windows 11	**Example:** Linux on a multi-core computer

Chapter 3.3: Windows Operating System

1 User Interface

A **User Interface (UI)** is the way through which a user communicates with and controls a computer. The two common types of user interfaces are **CUI/CLI** and **GUI**.

1.1 CUI / CLI (Character/Command Line User Interface)

CUI (Character User Interface) or **CLI (Command Line Interface)** allows the user to interact with the computer by **typing commands using the keyboard**.

For example, in Command Prompt, a user can type:

```
dir
```

to view files and folders.

Features of CUI/CLI

- Uses text-based commands.
- Mainly uses the keyboard.
- Requires knowledge of commands.
- Uses very few graphical elements.
- Can be fast for experienced users.
- Requires less memory and processing power.

Examples: MS-DOS and Windows Command Prompt.

1.2 GUI (Graphical User Interface)

GUI (Graphical User Interface) allows users to interact with the computer using **graphical elements** such as icons, menus, buttons, and windows.

For example, in Windows, a user can double-click an icon to open a program.

Features of GUI

- Uses icons, windows, menus, and buttons.
- Mainly uses the mouse and keyboard.
- Easy to learn and use.
- Does not require memorizing many commands.
- Provides a visual and user-friendly environment.
- Supports point-and-click operations.

Examples: Microsoft Windows and macOS.

1.3 Difference Between CUI/CLI and GUI

Basis	**CUI/CLI**	**GUI**
Interaction	Uses typed commands	Uses graphical elements
Input	Mainly keyboard	Mouse and keyboard
Ease of use	Difficult for beginners	Easy for beginners
Commands	Commands must be remembered	Commands are selected using menus and icons
Appearance	Text-based	Graphical and visual

2 Introduction to Windows OS

Windows is a popular **graphical operating system developed by Microsoft** that provides an easy-to-use environment for running programs, managing files and folders, and controlling computer hardware.

Some common versions of Windows include **Windows 7, Windows 10, and Windows 11**.

2.1 Features of Windows

Some important features of Windows are:

1. Provides a graphical and user-friendly interface.
2. Allows multiple applications to run at the same time.
3. Helps users create, organize, copy, move, and delete files.
4. Supports devices such as printers, keyboards, and USB drives.
5. Provides features such as passwords, user accounts, and security tools.

2.2 Main Elements of Windows

- **Desktop:** The main screen that appears after Windows starts.
- **Icons:** Small graphical symbols representing programs, files, folders, or other items.
- **Pointer:** The arrow or other symbol controlled by the mouse.
- **Taskbar:** The bar that shows open applications and provides quick access to programs.
- **Windows:** Rectangular areas where applications, files, or folders are displayed.
- **Start Button:** Provides access to applications, settings, files, and other Windows features.

3 Files and Folders

3.1 File

A **file** is a collection of related data stored on a computer with a specific name.
Examples:

- `notes.docx`
- `photo.jpg`
- `song.mp3`

Files are used to store different types of information such as text, images, videos, audio, and programs.

3.2 Folder

A **folder** is a container used to **store and organize files and other folders**.
For example, a folder named `College Notes` may contain different subject files:

```
College Notes
    |
    +-- Computer.docx
    +-- Mathematics.pdf
    +-- OperatingSystem.docx
```

Folders make it easier to organize, manage, and find files.

4 File Extensions

A **file extension** is the group of letters after the **dot (.)** in a filename. It tells the operating system what type of file it is and which program can usually open it.
For example:

```
assignment.docx
```

Here, `.docx` is the file extension.

4.1 Common File Extensions

Extension	**File Type**	**Example**
.txt	Text file	notes.txt
.docx	Microsoft Word document	report.docx
.pdf	PDF document	book.pdf
.xlsx	Excel spreadsheet	marks.xlsx
.pptx	PowerPoint presentation	presentation.pptx
.jpg / .jpeg	Image	photo.jpg
.png	Image	logo.png
.gif	Image/Animated file	animation.gif
.mp3	Audio file	song.mp3
.mp4	Video file	movie.mp4
.avi	Video file	video.avi
.zip	Compressed file	files.zip
.exe	Executable program	setup.exe
.html	Web page	index.html
.csv	Comma-separated data file	data.csv

Chapter 3.4: Open Source and Mobile Operating Systems

1 Open Source Operating System

An **Open Source Operating System** is an operating system whose **source code is freely available** for users and developers to view, modify, and distribute according to its license.

Open source operating systems are usually developed by a community of developers rather than by a single company.

Features of Open Source OS

- Source code is available to users and developers.
- Can be modified and customized.
- Usually has strong community support.
- Developers can find and fix problems.
- Often available free of cost.

Examples: Linux, Ubuntu, and FreeBSD.

1.1 Linux Operating System

Linux is an open-source operating system based on the Linux kernel. It is widely used on computers, servers, embedded systems, and many other devices.

Linux is known for its **security, stability, flexibility, and customization**. Users can modify the system according to their requirements.

Examples of use: Web servers, personal computers, supercomputers, and embedded devices.

Features of Linux

- Open source and highly customizable.
- Secure and stable.
- Supports multitasking and multi-user operation.
- Available in many distributions.
- Widely used for servers and development.

1.2 UNIX Operating System

UNIX is a powerful operating system originally developed at Bell Labs. It was designed to support multiple users and multiple tasks efficiently.

UNIX is widely known for its **stability, security, and powerful command-line environment**. Many modern operating systems and operating-system concepts have been influenced by UNIX.

Features of UNIX

- Supports multiple users.
- Supports multitasking.
- Provides strong security.
- Stable and reliable.
- Provides a powerful command-line interface.

Examples: UNIX systems are used in servers, workstations, and other large computing environments.

1.3 Linux Distribution: Ubuntu

Ubuntu is a popular Linux distribution based on the Linux operating system. It is developed and maintained by Canonical and a large open-source community.

Ubuntu provides a **user-friendly graphical interface** and is suitable for beginners as well as experienced users.

Features of Ubuntu

- Free and open source.
- Easy-to-use graphical interface.
- Secure and stable.
- Supports many software applications.
- Available for desktops and servers.
- Receives regular updates and security fixes.

Example: Ubuntu Desktop can be used as an alternative to Windows for everyday computer activities.

2 Mobile Operating System

A **Mobile Operating System (Mobile OS)** is an operating system designed specifically for **mobile devices** such as smartphones, tablets, and other portable devices.

It manages the device's hardware and provides an environment for running mobile applications.

Features of Mobile OS

- Provides a touch-based user interface.
- Manages mobile hardware and resources.
- Supports mobile applications.
- Provides wireless communication and networking.
- Manages features such as cameras, GPS, and sensors.
- Provides security and user privacy features.

3 Types of Mobile Operating Systems

3.1 Android OS

Android is a mobile operating system developed primarily by Google and based on the Linux kernel. It is widely used in smartphones, tablets, TVs, and other smart devices.

Example: Samsung, Google Pixel, and Xiaomi smartphones.

3.2 iOS

iOS is a mobile operating system developed by Apple for its mobile devices. It is known for its simple interface, security, and integration with Apple's hardware and services.

Example: iPhone.

3.3 Palm OS

Palm OS was a mobile operating system originally developed for personal digital assistants (PDAs). It was designed to provide a simple and easy-to-use interface for portable devices.

Example: Palm handheld devices.

3.4 BlackBerry OS

BlackBerry OS was a mobile operating system developed by BlackBerry for its smartphones. It was particularly known for its **secure communication and physical QWERTY keyboards**.

Example: BlackBerry smartphones.

3.5 Symbian OS

Symbian OS was a mobile operating system widely used in early smartphones. It was especially popular in Nokia mobile phones before the widespread adoption of Android and iOS.

Example: Nokia smartphones.

3.6 HarmonyOS

HarmonyOS is an operating system developed by Huawei. It is designed for smartphones and a wider range of connected and smart devices.

HarmonyOS focuses on providing a connected experience across different types of devices.

Example: Huawei smartphones and other Huawei smart devices.